

TiendaControl POS — Executive Business Concept & Structural Blueprint

Offline-First Point of Sale & Inventory Platform for Colombian Retail Markets

Executive Summary: TiendaControl POS is an offline-first, low-latency POS and inventory management ecosystem tailored for Colombian *tiendas de barrio* and *minimercados*. It addresses critical operational pain points—cash leakage, unmanaged customer credit ("fiado"), and stockouts—while maintaining compliance with local DIAN electronic invoicing regulations.

Phase 1: Operational Pain Points & Module Specifications

Local retail markets in Colombia face systemic revenue loss due to non-digitized workflows. The following software modules target these operational vulnerabilities directly:

Pain Point	System Feature	Business Outcome
1. Internal Theft & Cash Leaks	Blind Cash Audits (<i>Cuadre Ciego</i>) & Petty Cash Logging	Eliminates shift discrepancy manipulation and forces physical cash accountability.
2. Untracked Credit ("Fiado")	Digital Credit Ledger with WhatsApp Integration	Reduces debt recovery cycles and enforces automated credit spending caps.
3. Stockouts & Expiration Waste	Dynamic Reorder Points & Expiration Dashboards	Prevents stockout revenue loss and minimizes expired inventory write-offs.

1.1 Cash Reconciliation & Shift Controls

- Blind Cash Audits (*Cuadre Ciego*):** At shift balance, the system requires cashiers to enter physical counts per bill denomination without displaying expected remote backend totals.
- Petty Cash Logging (*Caja Menor*):** Every mid-shift cash extraction (e.g., paying supplier deliveries at the door) requires a selected category, timestamp, and mandatory PIN confirmation.
- Automated Threshold Alerts:** Discrepancies exceeding defined limits generate an instant alert flag for store management.

1.2 Digital Credit Ledger (Control de Fiado)

- **Customer Accounts:** Links accounts directly to national identity cards (CC) or mobile phone numbers.
- **Automated Credit Caps:** Enforces account-level maximum balance limits, automatically blocking further credit transactions when limit or age thresholds are exceeded.
- **WhatsApp Statement Dispatch:** Generates concise balance summaries and payment requests sent via WhatsApp link in a single tap.

1.3 Inventory Intelligence & Margin Tracking

- **Automated Reorder Alerts:** Triggers notifications when stock reaches defined safety thresholds per product.
- **Batch & Expiration Tracking:** Monitors expiration windows (7, 15, and 30-day horizons) to optimize item rotation.
- **Margin Analysis:** Calculates real-time margins against acquisition costs to preserve profitability during inflationary adjustments.

Phase 2: Technical Architecture & System Infrastructure

2.1 Hardware Integration Layer

- **Barcode Scanning (HID Buffer):** A root-level keyboard event buffer captures rapid keystroke sequences (<50ms) from USB/Bluetooth handheld scanners without requiring active UI element focus.
- **Optical Fallback:** Integrates camera-based barcode scanning for inventory counts and mobile checks.
- **Thermal Printing & Cash Drawers:** Utilizes ESC/POS protocol via Bluetooth, USB, or local network (LAN) for standard 58mm/80mm receipt printers, sending RJ11 signals to trigger physical cash drawers on completed transactions.

2.2 Offline Synchronization Engine (Event-Sourced Ledger)

To operate reliably without active internet, the client runs on a local embedded SQLite database. Rather than attempting full-state synchronization (which introduces data race conditions during concurrent offline writes), the system employs an **Event-Sourced Delta Ledger**.

- **Atomic Event Logging:** Every state change is stored as a immutable local event payload (e.g., SALE_CREATED, CASH_OUT, STOCK_ADDED) paired with precise client timestamps and unique UUIDs.
- **Server Reconciliation:** Upon network restoration, client deltas stream to the cloud repository (PostgreSQL). The server processes events sequentially per tenant, computing accurate inventory states without state overwrite loss.

2.3 DIAN Regulatory Compliance (Resolution 000165)

- **Dual-Receipt Mechanism:** Generates an instant local POS receipt for the customer offline.
- **Asynchronous Electronic Invoicing:** Queues XML generation, attaches digital cryptographic signatures, and transmits data to DIAN endpoints automatically upon network reconnection.

Phase 3: Business Model & Execution Strategy

3.1 SaaS Pricing Structure

Tier	Target Segment	Monthly Price (COP)	Core Features
Básico	Small <i>Tiendas de Barrio</i>	\$40,000 – \$60,000	1 Device, Offline POS, Blind Cash Audits, Barcode Search, Local Inventory.
Pro	Medium <i>Minimercados</i>	\$90,000 – \$120,000	Multi-Device Sync, Credit Ledger with WhatsApp, DIAN POS Integration, Expiration Alerts.
Hardware Bundle	Turnkey Setup	One-time Purchase	Pre-configured Android Tablet/Terminal + Thermal Printer + Handheld Scanner.

3.2 Go-To-Market & Onboarding Acceleration

- **Field Operations:** Direct outreach in commercial hubs and wholesale centers.
- **Pre-Loaded FMCG Catalog:** Provides a default catalog containing Colombia's top fast-moving consumer goods (standard barcoded items and brand packaging) to eliminate manual entry friction during setup.

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